Answer key: Lesson 8, Make a Table

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Squares in a row are made with sticks. Look at the three pictures. Count the sticks in each.

Answer: 4; 7; 10

The first square has 4 sticks. The second square shares one side, so it adds 3 more sticks: 7 in all. The third square adds 3 more: 10 in all.

Watch for: Writing 8 for 2 squares (4 + 4): the squares share a side.; Writing 12 for 3 squares.

S2. For every 3 red beads there are 2 blue beads. Fill in the table. The last column is spare.

Answer: 9 and 8; 15

Red beads go up by 3 each time: 3, 6, 9, 12. Blue beads go up by 2 each time: 2, 4, 6, 8. 10 blue beads is 5 groups, and 5 groups have 15 red beads.

Watch for: Adding 3 to the blue beads, or 2 to the red beads.; Writing 10 or 12 for the red beads.

S3. Siti keeps adding squares to the row. Fill in the table. The last column is spare.

Answer: 13, 16, 19; 3; 22

The sticks go up by 3 each time, because each new square shares a side: 4, 7, 10, 13, 16, 19. 7 squares: 19 and 3 more is 22 sticks.

Watch for: Adding 4 each time.; Stopping at 19 for 7 squares (that is 6 squares).

S4. Ravi reads 5 pages each day. The table shows the pages he has read in all. Fill in the table.

Answer: 15, 25, 30; day 5

Each day adds 5 pages: 5, 10, 15, 20, 25, 30. 25 pages in all is day 5.

Watch for: Writing 5 again in each box.; Adding 4 or 6 instead of 5.

S5. Look at the squares in a row. Tick True or False for each statement.

Answer: true; false; true

Each new square shares a side, so it adds 3 sticks: True. 4 squares need 13 sticks, not 16: False. 6 squares need 19 sticks: True.

Watch for: Saying 4 squares need 4 times 4 = 16 sticks.; Not checking the statement against the table.

Think harder (page 3)

T1. Mi makes triangles in a row with sticks. Fill in the table. Then finish the rule.

Answer: 21 and 41; add 1

From 4 to 10 triangles there are 6 more triangles, and each adds 2 sticks: 9 and 12 make 21. Or double 10 is 20, and 1 more is 21. For 20 triangles: double 20 is 40, and 1 more is 41. The rule is: double the number of triangles, then add 1.

Watch for: Writing 30 or 60 (3 sticks for every triangle).; Writing 11 for the 10th picture (counting on only one step).

T2. Look at rows of triangles made with sticks.

Answer: No; odd; Yes

Each row has double the triangles and then 1 more, so the number of sticks is an even number plus 1, which is odd. 20 is even, so no row of triangles uses exactly 20 sticks. 25 is odd: 25 take away 1 is 24, and half of 24 is 12, so 12 triangles use exactly 25 sticks.

Watch for: Saying Yes after finding a close number such as 21.; Thinking that every number of sticks is possible.

T3. A child made a table for squares in a row. One number is wrong.

Answer: 4; 13; 3

The numbers go up by 3 each time: 4, 7, 10, then 13, 16, 19. The number for 4 squares is 12, but it should be 13.

Watch for: Choosing 5 because 16 comes after the wrong number.; Writing 11 or 14.

Go further (page 4)

G1. Picture 100. Look at the tables for rows of triangles and rows of squares.

Answer: 201; 301; 151

Triangles: double 100 is 200, and 1 more is 201. Squares: the first square has 4 sticks and each of the other 99 squares adds 3: 99 times 3 is 297, and 4 more is 301. Or 100 times 3 is 300, and 1 more is 301. 50 squares: 50 times 3 is 150, and 1 more is 151.

Watch for: Writing 300 or 400 (3 or 4 sticks for every picture).; Writing 200 for the triangles and forgetting the 1.

G2. Work back from the number of sticks.

Answer: 15; 12

Triangles: 31 minus 1 is 30, and half of 30 is 15. Squares: 37 minus 1 is 36, and 36 shared into groups of 3 is 12.

Watch for: Dividing 31 by 3 for the triangles.; Forgetting to take away the 1 first.

G3. A row of squares and a row of triangles have the same number of pictures.

Answer: 10; 20; 5

10 squares need 31 sticks and 10 triangles need 21 sticks: 10 more. 20 squares need 61 sticks and 20 triangles need 41 sticks: 20 more. 5 squares need 16 sticks and 5 triangles need 11 sticks: 5 more. Each new picture needs 1 more stick for a square than for a triangle.

Watch for: Comparing the first pictures only (4 and 3).; Subtracting 10 and 20 from each other.

Investigation (page 5): Strips left over

First: 7 strips. The most triangles is 6 (13 strips, 1 left over). The most squares is 4 (13 strips, 1 left over).
No: a row of triangles uses an odd number of strips, so 14 never works, and the squares use 4, 7, 10, 13 or 16 strips, so 14 does not work either.

Look for: A child who builds with the strips and counts; a child who uses the table to find how far 14 goes; a child who says why 14 never fits (odd numbers for triangles; 13 then 16 for squares). Credit any clear reason. Do not mark a child down for stopping at one shape.

Exit ticket (page 5)

E1. Each tray holds 7 cups. 1 tray has 7 cups. 2 trays have 14 cups.

Answer: 21

Each tray adds 7 cups: 7, 14, 21. 3 trays have 21 cups.

Watch for: Writing 20 or 28.; Writing 3 (the number of trays).

E2. Mei reads 3 pages each day. How many pages does she read in 5 days? Choose the correct answer. Write its number in the bracket.

Answer: (3) 15

Each day adds 3 pages: 3, 6, 9, 12, 15. In 5 days she reads 15 pages.

Watch for: Choosing (1) 8, which adds 3 and 5.; Choosing (4) 35.

E3. Rows of triangles use 3, 5, 7 and 9 sticks for 1, 2, 3 and 4 triangles. Each new triangle needs 2 more sticks.

Answer: 17

Go on in twos: 4 triangles 9, 5 triangles 11, 6 triangles 13, 7 triangles 15, 8 triangles 17. Or double 8 is 16 and 1 more is 17.

Watch for: Writing 24 (8 times 3): the shared sticks are counted twice.; Stopping early at 15 (7 triangles) or 16.

Sneak-peek puzzles (not in the workbook)

1. 12 pencils
2. 14 dots. Each picture has 3 more dots than the one before: picture 4 has 11 and picture 5 has 14.